MICHIGAN MEDICINE
UNIVERSITY OF MICHIGAN

Guidelines for Treatment of Odontogenic Infections in Hospitalized Adults

Clinical Setting	Empiric Therapy	Duration	Comments
Suppurative (pyogenic) orofacial odontogenic infection, including: • **Acute apical periodontitis** • **Acute dentoalveolar abscess** • **Space infection around the face (local extension depends on the tooth involved):** - **Masticator space** - **Buccal space** - **Canine space** - **Parotid space** - **Submandibular space** - **Submental space** - **Vestibular space** - **Ludwig's angina** • **NOT including deep head and neck infection** Pathogens: *Streptococcus viridans* *Streptococcus anginosus* *Peptostreptococci* *Prevotella* *Fusobacterium* *Porphyromonas* *Bacteroides* spp. *Veilonella* *Actinomyces* *Propionobacterium* *Capnocytophaga* Other uncommon pathogens: *Staphylococci* spp Enteric Gram negative bacilli	1st line: **Ampicillin-sulbactam** 3 g IV q6h* Low-to-High Risk PCN Allergy**: **Cefazolin** 2 g IV q8h* + **Metronidazole** 500 mg IV/PO q8h Severe non-IgE PCN Allergy**: **Levofloxacin** 750 mg IV q24h* + **Metronidazole** 500 mg IV/PO q8h	For acute apical periodontitis and acute dentoalveolar abscess: In the presence of surgical control: 5 days post drainage In the absence of surgical control: Duration is dependent on clinical and/or radiographic improvement. Minimum of 7 days AND at least 3 days of clinical improvement Ludwig's angina: 3 weeks	• **The most important element is surgical drainage and removal of necrotic tissue.** • Blood cultures should be sent when systemic signs are involved • If abscess is drained, aerobic and anaerobic bacterial cultures should be sent. • ***Strep anginosus*, a prominent pathogen in these infections, is resistant to clindamycin >25% of the time it is isolated in our hospital.** • Consider ID consult for Ludwig's angina case • Coverage for *Actinomyces* may be considered in extensive infections, which would affect both coverage choices and duration. Oral step-down options: • 1st line: **Amoxicillin-clavulanate** 875 mg PO BID* • Low-to-High Risk PCN Allergy**: **Cefuroxime** 500 mg PO BID* + **Metronidazole** 500mg PO TID • Severe non-IgE PCN Allergy**: **Levofloxacin** 750 mg PO daily* + **Metronidazole** 500 mg PO TID

Clinical Setting	Empiric Therapy	Duration	Comments
Suppurative (pyogenic) orofacial odontogenic infection in: **1) Severely immunocompromised patients** **2) Patients who have severe sepsis and/or septic shock** **3) Patients who had in-hospital surgical procedure in the past 90 days** Pathogens: *Streptococcus viridans* *Streptococcus anginosus* *Peptostreptococci* *Prevotella* *Fusobacterium* *Porphyromonas* *Bacteroides* spp *Veilonella* *Actinomyces* *Propionobacterium* *Capnocytophaga* *Staphylococci* spp Enteric Gram negative bacilli including *P. aeruginosa*	1st line: **Vancomycin** IV (see nomogram)* + **Piperacillin-Tazobactam** 4.5 g IV q6h Low-to-High Risk PCN Allergy**:: **Vancomycin** IV (see nomogram)* + **Cefepime** 2 g IV q8h* + **Metronidazole** 500 mg IV/PO q8h Severe non-IgE PCN Allergy**: **Vancomycin** IV (see nomogram)* + **Aztreonam** 2 g IV q8h* + **Metronidazole** 500 mg IV/PO q8h	For acute apical periodontitis and acute dentoalveolar abscess: Duration is dependent on surgical debridement, clinical & radiographic improvement. Minimum of 7 days AND at least 3 days of clinical improvement Oral stepdown therapy depends on clinical improvement and microbiologic data. Ludwig's angina: 3 weeks	• Severely immunocompromised patients: neutropenia, allogeneic HSCT, HIV accompanied by CD4 <200 • **The most important element in surgical drainage and removal of necrotic tissue.** • ID consult is recommended • Blood cultures should be sent when systemic signs are involved • If abscess is drained, aerobic and anaerobic bacterial cultures should be sent. • ***Strep anginosus*, a prominent pathogen in these infections, is resistant to clindamycin >25% of the time it is isolated in our hospital.** • Coverage for *Actinomyces* may be considered in extensive infections, which would affect both coverage choices and duration.
Mandibular Osteomyelitis Pathogens: *Streptococcus viridans* *Streptococcus anginosus* *Peptostreptococci* *Prevotella* *Fusobacterium* *Porphyromonas* *Bacteroides*spp *Veilonella* *Actinomyces* *Propionobacterium* *Capnocytophaga* Other uncommon pathogens: *Staphylococci* spp. Enteric Gram negative bacilli *Candida* spp.	Consider holding antibiotics until bone cultures can be obtained in hemodynamically stable patients 1st line: **Ampicillin-sulbactam** 3 g IV q6h* Low-to-High Risk PCN Allergy**: **Ceftriaxone** 2 g IV q24h + **Metronidazole** 500 mg IV/PO q8h Severe non-IgE PCN Allergy**: **Moxifloxacin** 400 mg IV/PO q24h (see comment#) If mandibular osteomyelitis is secondary to contiguous spread of exposed bone from Osteoradionecrosis leading to the skin, then would recommend the addition of **vancomycin** to empiric therapy.	Final regimen pending microbiologic data. Duration to be determined by clinical improvement and serial evaluation, Typically 6 weeks.	• ID consult strongly recommended. • When osteomyelitis is suspected, it is advised to attempt surgical debridement of necrotic bone, and **to send purulence and bone for pathology as well as anaerobic bacterial, aerobic bacterial and *Actinomyces* culture to help guide therapy.** • **Strep anginosus, a prominent pathogen in these infections, is resistant to clindamycin >25% of the time it is isolated in our hospital.** • In the setting of mandibular osteomyelitis caused by tooth extraction or odontogenic infection, the typical oral flora are expected pathogens. • # Moxifloxacin provides robust coverage against most oral flora but fluoroquinolones have variable activity against *Actinomyces* and are unproven clinically. As such, consideration should be given to adding Doxycycline when there is a high clinical suspicion for *Actinomyces*.

* Dose may need to be adjusted for renal dysfunction

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- See Beta-lactam Allergy Evaluation and Empiric Therapy Guidance document for further information on safe treatment strategies in patients with purported β-lactam allergies.
- Consult the Beta-Lactam Allergy Evaluation Service (BLAES) to assess patients with purported β-lactam allergies
- Low risk includes remote unknown reaction, pruritis without rash, urticaria/hives with no other symptoms, and mild to severe rash with no other symptoms.
- High risk includes:
    1. Anaphylaxis
    2. Cardiovascular symptoms (syncope/passing out, arrhythmia)
    3. Angioedema/swelling
    4. Respiratory distress
- Severe non-IgE mediated PCN allergy includes organ damage (kidney/liver), drug-induced immune-mediated anemia/thrombocytopenia/leukopenia, Stevens-Johnson Syndrome/Toxic Epidermal Necrosis, rash with pustules, DRESS, serum sickness

Last revised: 06/2024	**Antimicrobial Stewardship Subcommittee:** 06/2024	**P&T:** 07/2024

References	
	Citation

Revision History	
Date	**Revision**
03/2022	Updated vancomycin dosing & hyperlinks
06/2024	Updated allergy wording

The recommendations in this guide are meant to serve as treatment guidelines for use at Michigan Medicine facilities. If you are an individual experiencing a medical emergency, call 911 immediately. These guidelines should not replace a provider's professional medical advice based on clinical judgment, or be used in lieu of an Infectious Diseases consultation when necessary. As a result of ongoing research, practice guidelines may from time to time change. The authors of these guidelines have made all attempts to ensure the accuracy based on current information, however, due to ongoing research, users of these guidelines are strongly encouraged to confirm the information contained within them through an independent source